

A conditional field model of cortical coherence and recovery

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Abstract

How could changes in the physical coupling of neural populations constrain the recovery of coherent activity after sleep? We develop a conditional field model connecting information dissipation, cortical coupling, and compatible local states. A Lean 4 formalization separates established mathematical results from eight explicit assumptions needed to compose them; a common toy model demonstrates their joint satisfiability without identifying them with cortical physiology. The experimentally relevant component is a noisy mean-field phase model in which a slowly varying coupling crosses a synchronization threshold. Its stationary coherent branch has square-root onset and a von Mises concentration–coherence relation. Finite simulations delimit these predictions: ramp speed changes the apparent onset exponent, spatial coherence depends on the specified coupling regime, and coupling plasticity preserves coherence without recovering the tested environmental structure. An exploratory scalp-EEG analysis is compatible with the concentration–coherence relation over a range too narrow to distinguish it from a linear approximation. The framework therefore supplies a testable hypothesis for coupling-gated recovery, rather than a derivation of consciousness. Its decisive requirements are independent calibration of effective cortical coupling and an awakening experiment that separates slow coupling changes from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the proposed collective state remains an empirical question.

We ask what a field account must assume to connect physical constraints on information processing to a coherent, globally consistent state. The model combines finite information capacity, thermodynamic bounds on memory, a continuum coupling description, synchronization, and the gluing of compatible local states. A reflexive extension requires the resulting state to be a fixed point of a map that reads its own local encoding. These are distinct constructions: thermodynamic efficiency does not imply synchronization, and synchronization does not imply either compatible representations or subjective experience.

The contribution has three parts. First, a machine-checked composition makes the missing connections explicit. Second, a slowly varying coupling yields a conditional prediction for recovery after awakening, with a stationary

23 square-root onset and a specified phase-distribution relation. Third, numer-
24 ical controls test the robustness and limits of the dynamical interpretation,
25 including negative results. The established ingredients include Landauer’s
26 bound (Landauer, 1961), the dissipation cost of nonpredictive information
27 (Still et al., 2012), the noisy Kuramoto transition (Sakaguchi, 1988), and
28 Banach’s fixed-point theorem (Banach, 1922). Their biological composition,
29 rather than any new fundamental physical law, is the object of study.

30 The proposed connection to consciousness is restricted. We use a global
31 section to represent consistency across local descriptions and a reflexive fixed
32 point to represent self-consistency. Identifying either construction with phe-
33 nomenal unity is a philosophical commitment, not a theorem or an outcome
34 of the simulations. The biological tests concern coupling, coherence and
35 recovery; they can challenge the physical model without resolving that iden-
36 tification.

2 A coupling-gated model of cortical coherence

2.1 Dynamics and stationary solutions

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles treated as dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters; a frequency-spread width cannot be substituted for D when applying the identical-frequency stationary theory.

For identical frequencies in their rotating frame, the assumed stationary density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

The I_n are modified Bessel functions, and ψ is the collective phase. Lean verifies normalization, the density's circular order parameter, and the self-consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent order parameter is zero. At $K > K_c$, exactly one positive solution coexists with zero; the positive branch is continuous at threshold and strictly

53 increasing with K .

54 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

55 This is a result about the stationary solution branch. Of the coherent branch
56 we prove existence, uniqueness, continuous emergence at threshold and strict
57 growth in K , but not that it is dynamically selected. The dynamical mean-
58 field limit—propagation of chaos—is the missing link between the finite sys-
59 tem and the stationary description (Sznitman, 1991). No theorem in this
60 development closes it. Exchange symmetry alone does not justify factoriza-
61 tion of the joint law, particularly for an ensemble with an unpinned collective
62 phase.

63 A spatial version replaces the finite sum by an integral against a kernel
64 $K(x, y)$. We normalize the substrate measure to total mass one, so the
65 mean-field strength is a kernel average rather than a quantity that grows
66 with the arbitrary measure of the substrate. Reduction of a heterogeneous
67 spatial kernel to a scalar K is an additional modelling step. The threshold
68 in Eq. (2) does not automatically apply to every spatial or signed network.

69 **2.2 Physical identification and calibration**

70 Regular mesh refinement gives convergence of a discrete coupling-energy
71 quadrature to its continuum integral. This is scalar energy convergence,

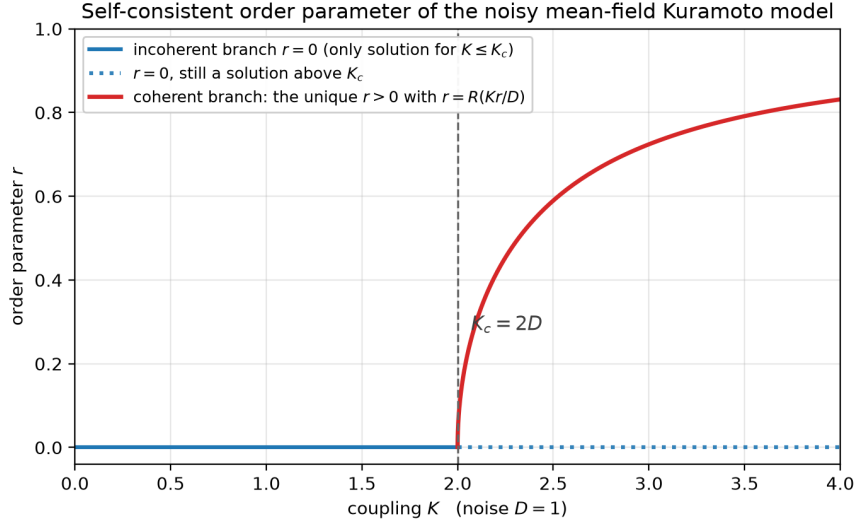


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

not convergence of trajectories and not a construction of a physical kernel.
The model posits a continuum kernel and separately identifies its cortical
realization with endogenous electromagnetic coupling. The formal predicate
requires a jointly continuous kernel, a normalized substrate, a mean-field
strength equal to its spatial average, and no single site providing more than
half of that mean-field coupling. Toy witnesses satisfy these conditions; none
is a model calibrated to cortex.

Extracellular fields can modulate spike timing (Fröhlich & McCormick,
2010; Anastassiou & Koch, 2015), and field-based models have been proposed

81 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
82 Fadden, 2020). These observations motivate an effective interaction. They
83 do not establish its magnitude relative to phase diffusion, its spatial averag-
84 ing rule, or its causal role in conscious experience. Likewise, changes in phase
85 organization under propofol (Bastos et al., 2021) do not measure a crossing
86 of the model’s coupling threshold.

87 The physical-unit conversion is a material limitation. An estimate pro-
88 portional to $N\Delta tf$, using population size, a spike-time shift and oscillation
89 frequency, is dimensionless. It cannot be compared directly with an inverse-
90 time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

91 requires a rate calibration γ , where s is the time shift per unit field and E the
92 field amplitude. The saved field-strength calculations are conditional on this
93 calibration and on the chosen spatial aggregation; they do not independently
94 establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction kernel and phase
95 diffusion from the same physical system is necessary before the threshold can
96 become a quantitative biological claim.

3 Recovery after awakening: a conditional prediction

Sleep-associated changes in extracellular space and clearance motivate a possible slow control of coupling (Xie et al., 2013). We hypothesize that recovery after awakening includes a change in effective coupling $K(t)$, mediated by extracellular geometry, on a slower timescale than phase dynamics. Neither the monotonic relation between geometry and coupling nor its role in human sleep inertia follows from the cited physiological observation. The model requires both to be measured.

If $K(t)$ rises through K_c and the phase distribution tracks the coherent stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

The ideal stationary description has a subthreshold interval followed by continuous onset. Saturation requires a specified subsequent coupling trajectory; monotonic coupling alone does not determine a logistic or any unique sigmoid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently known. When coupling is inferred from the same coherence trace, this is not a parameter-free prediction of latency.

Critical slowing down limits the adiabatic prediction in Eq. (5) near threshold. Finite populations also have nonzero fluctuating order below

threshold. Consequently, an observable onset requires a predeclared threshold above the finite-size floor, and a finite-rate trajectory need not display an infinite-slope cusp. The ramp controls below quantify this limitation.

3.1 Distributional and temporal tests

The stationary model makes a second prediction: phase concentration and coherence obey $r = I_1(a)/I_0(a)$. This relation has no fitted shape parameter, but its applicability depends on the stationary-density assumption and the observation model. It is common to the von Mises family and therefore cannot by itself identify ephaptic coupling or distinguish this framework from every alternative synchronization model. Within the same ansatz, $a/r = K/D$ for $r > 0$; the limiting threshold ratio is a consequence of the density relation, not an independent test of it. Ratios near the finite-size floor are especially unstable.

The estimators must be independent. Estimate r by the circular resultant length and a by an unweighted regression of log-density on the cosine of phase relative to the circular mean. Estimating a by inverting the Bessel ratio from r would enforce the relation being tested. Neither estimator should use linear statistics on wrapped phases. At least 100 spatial samples are required by the implemented density estimator; with fewer sites, pooling over time changes the estimand to a spatiotemporal distribution and must be stated explicitly.

An awakening experiment should combine phase estimates, an independent measure of recovery performance, and a separately validated indicator

138 of extracellular geometry. A diffusion-MRI observable or an astrocytic signal
 139 is not itself a measurement of K ; its relation to effective coupling requires
 140 calibration. The analysis should compare coupling-gated recovery with ex-
 141ponential, flexible sigmoidal and other slow-recovery models under the same
 142temporal filtering and observation noise. Latency alone cannot distinguish
 143the mechanism, because multiple processes can create delays.

144 Useful outcomes are failure of the stationary phase-density relation over
 145a sufficiently broad concentration range, absence of the proposed coupling-
 146geometry association, or a recovery trace better explained by alternatives
 147after finite-rate and observation effects are accounted for. A linear appar-
 148ent onset is evidence against the adiabatic approximation only where that
 149approximation is independently justified. It does not falsify the whole field
 150hypothesis under arbitrary ramp speeds. Similarly, the absence of a fold
 151in the stationary equation excludes stationary coexistence of two positive
 152branches in this model; it does not exclude finite-rate hysteresis or adjudi-
 153cate all theories of ignition.

154 **4 Thermodynamic constraints and the for-** 155 **mal composition**

156 The synchronization model is embedded in a broader conditional construc-
 157tion. Table 1 lists its eight connecting hypotheses. Four are recorded as
 158formalization gaps, two as modelling assumptions and two as physical com-

159 mitments. These categories describe their role in the composition, not their
160 plausibility or ease of proof. In particular, finite capacity does not imply a
161 symmetry-breaking boundary: these are independent starting assumptions,
162 even though the composition can accept an implication between them as an
163 argument.

Edge	Status	Required connection
E12	Formalization gap	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
E23	Formalization gap	The boundary is associated with a finite register whose update leaves states unreachable.
E34	Formalization gap	Erasure supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45	Modelling assumption	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56	Physical commitment	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Formalization gap	Coherence supplies a cover with coupling at least the named mean-field value, reached by relaxation and carrying compatible local states.
E89	Modelling assumption	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

164 4.1 Capacity, boundaries and predictive memory

165 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
 166 equality only for the uniform distribution. A stream with more possible
 167 histories than $|X|$ cannot be encoded injectively in that register. Separately,
 168 a continuous field on a connected substrate that reaches distinct components
 169 of a disconnected vacuum manifold must leave that manifold somewhere,
 170 the simplest topological-defect construction (Kibble, 1976). Relating this
 171 locus to a particular finite information-processing register is an additional
 172 hypothesis.

173 A non-surjective update on a finite register is non-injective. Under the
 174 specified system–bath model and thermal assumptions, erasure carries a Lan-
 175 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 176 update erases, and logical many-to-one processing must be distinguished from
 177 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 178 tion records the physical heat identity as an obligation of each model and
 179 supplies an explicit one-bit eraser satisfying it.

180 For a system state X_t driven by an environmental signal S_t , the predictive
 181 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (6)$$

182 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data processing
 183 makes the bracket nonnegative (Still et al., 2012). The bound constrains

184 nonpredictive memory at a given dissipation budget. It does not specify a
 185 learning rule, maximize prediction, or require the internal representation to
 186 reproduce environmental geometry. Connecting this budget to Landauer heat
 187 additionally requires nonpredictive information to be bounded by the entropy
 188 erased by the register. The formal bridge derives Eq. (6) from Landauer’s
 189 bound under that identification; it does not derive the identification.

190 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
 191 side tends to zero. Gibbs’ inequality proves nonnegativity of relative entropy,
 192 not a general identification of relative entropy with physical entropy produc-
 193 tion. These distinctions prevent an informational inequality from being read
 194 as a derivation of environmental learning, Bayesian inference, or semantic
 195 content.

196 4.2 Compatible states and reflexive fixed points

197 A sheaf formalizes local descriptions and their restriction to smaller regions.
 198 If local sections agree on every overlap, the sheaf condition gives a unique
 199 global section. In the probability-sheaf construction, this is consistency of
 200 local density information; it is not implied by agreement of oscillator phases.
 201 The development includes a counterexample with equal phases but incom-
 202 patible densities on an overlap. The model must exclude such configurations
 203 through its compatibility hypothesis.

204 A thermodynamic cover couples this gluing construction to a phase con-
 205 figuration at a potential minimum. For the deterministic positive-coupling

206 model, convergence to that minimum is proved under an explicit initial basin
207 condition. Other equilibria prevent an unconditional conclusion. The sta-
208 tionary noisy order parameter, the deterministic relaxation, and the cover
209 must therefore be connected by E78 rather than identified without a hypoth-
210 esis. Neither field continuity nor synchronization alone guarantees a globally
211 compatible representation.

212 A reflexive extension equips global sections with a complete metric and a
213 predictive map formed by restricting a state to a local encoding region and
214 reading that encoding out. If the map contracts and fixes the section supplied
215 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
216 The contraction rate is assumed, and the equality between the glued section
217 and the fixed point is part of E89. A coherent order parameter supplies the
218 supercritical inequality required by the stated contraction bound; it does not
219 construct the map or prove that cortical dynamics implements it.

220 The theorem `chain` concludes `UnifiedSelf` from these hypotheses. A
221 common toy witness satisfies all eight, so their conjunction is not empty.
222 The project declares no additional Lean axioms beyond the foundational
223 axioms used by Lean’s libraries; physical content resides in hypotheses and
224 model fields. This checks the conditional mathematical composition, not the
225 empirical truth of its premises. Detailed constructions, scope statements and
226 counterexamples are retained in the supplement and source files.

227 **5 Numerical controls and exploratory obser-** 228 **vations**

229 The simulations provide finite-regime evidence and do not discharge a formal
230 edge. Each new control uses explicit seeds and compact saved summaries; the
231 numerical values below are generated from those artifacts. The supplement
232 gives parameters, estimators, failure criteria and additional panels. Identical-
233 frequency controls use phase diffusion rather than a quenched frequency-
234 spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

235 **5.1 Finite-rate and mean-field controls**

236 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
237 four coupling speeds. Escape is the first of three consecutive saved samples
238 at $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest win-
239 dow. The three slower ramps have mean coupling delays 0.1947, 0.0684 and
240 0.0253, with a fitted delay exponent 0.443. The apparent onset exponent
241 changes from 0.964 at the fastest speed to 0.417 at the slowest. Thus the
242 measured onset depends on ramp speed; the stationary square-root law can-
243 not be imposed on an arbitrary recovery trace. The specified deviation-based
244 critical ramp speed is not bracketed by the sweep.

245 The dynamical-selection control shows escape toward a coherent branch
246 above threshold, but retains quantitative finite-size discrepancies. At $K =$
247 2.8, steady mean order is 0.67735 against the static value 0.68270. The

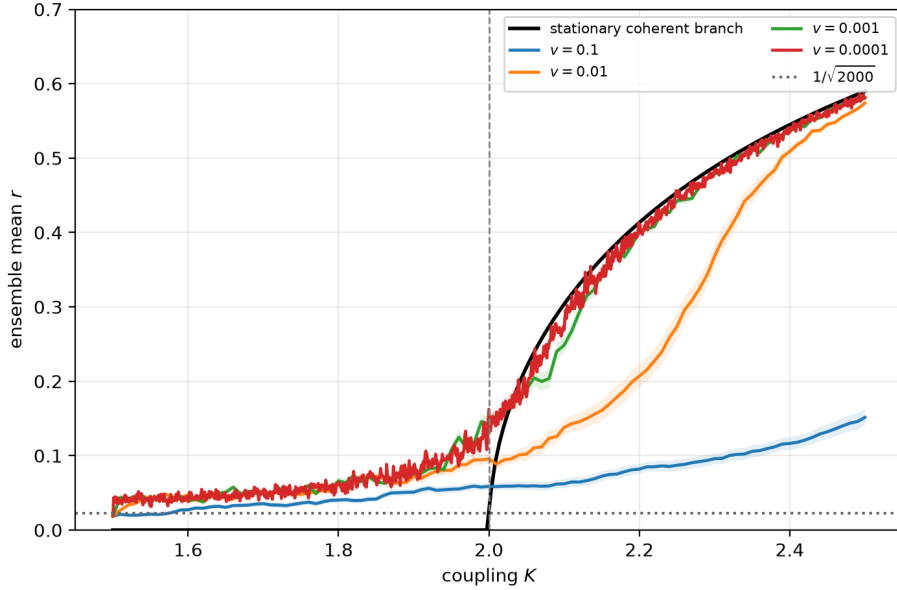


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

248 fitted growth-rate slope is 0.43240, compared with the theoretical slope $1/2$.
 249 The propagation-of-chaos diagnostics show decreasing subcritical dependence
 250 with increasing population size. In the supercritical unpinned ensemble,
 251 however, pair correlation remains substantial because realizations share a
 252 random collective orientation. This negative unconditional result does not
 253 settle a conditional mean-field limit after removal of that orientation.

254 **5.2 Spatial coupling, frustration and calibration**

255 A periodic two-dimensional sheet with row-normalized exponential coupling
256 remains coherent in the sampled empirical decay-length band under the de-
257 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
258 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
259 eter regime; fixing total coupling while changing range tests fragmentation,
260 not whether cortex supplies that total coupling.

261 A row-balanced Dale-law network shows that row balance alone does not
262 guarantee a floor-level incoherent baseline. In a separately specified weak-
263 synapse regime, a uniform modulatory term produces an operational co-
264 herence crossing at K_{eff}/D between 1.8994 and 1.9387. The physical field
265 conversion remains conditional on the rate calibration γ . Neither the spa-
266 tial control nor the rescue sweep measures cortical coupling or resolves the
267 unit-conversion limitation.

268 **5.3 Joint phase and coupling plasticity**

269 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (7)$$

270 It is a quadratic functional of an affine image of the kernel. The formalization
271 constructs the bounded drift operator A and its adjoint on the relevant L^2
272 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$. The specified gradient flow

decreases this objective at fixed phases. The objective is not generally a thermodynamic entropy-production rate, which requires probability-current information. Whether biological couplings follow this flow is an additional modelling question.

The joint control evolves noisy phases with clustered frequencies and nonzero grand-mean drive, while updating symmetric couplings every 50 fast steps. Couplings remain nonnegative, hollow and normalized to a fixed total. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so both endpoints contribute. Resource constraints and evolving phases remove the fixed-phase monotonicity guarantee.

Across three seeds, second-half order lies between 0.9811 and 0.9815. Matched frozen-coupling controls maintain similar coherence. The distance to the true cluster template increases in every adaptive run, with worse changes than the shuffled-label comparison. The template is a covariance of shared latent cluster signals, not an input to the update or a measured covariance of the constant frequencies. This regime therefore supplies no evidence that squared-drift plasticity learns environmental structure. Its positive objective plateau does not establish stationarity of the adaptive process or a thermodynamic entropy-production rate.

5.4 Exploratory EEG check

An analysis of public propofol-sedation EEG tests compatibility with the concentration-coherence relation (Bajwa et al., 2025). The implemented es-

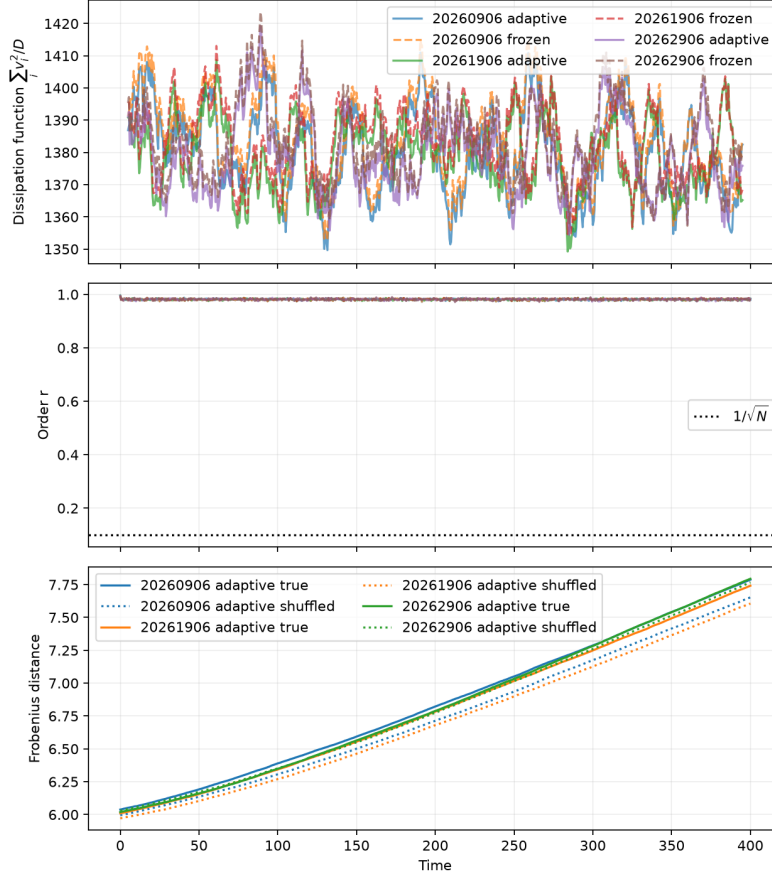


Figure 3: Joint fast/slow dynamics with matched frozen-coupling controls. Top: smoothed squared-drift dissipation, with frozen controls dashed. Middle: order and the finite-size reference. Bottom: distances to true cluster templates (solid) and node-label shuffles (dotted). Coherence persists, but structural distances increase. The result limits the proposed connection between dissipation minimization and environmental structure.

295 timator pools 62 channels over a 100-ms window; it therefore estimates a spa-
296 tiotemporal distribution rather than an instantaneous spatial one. Bipolar
297 and common-reference montages give different residuals, making observation
298 and referencing choices material to interpretation.

299 The bipolar results are compatible with the Bessel relation over the ob-
300 served concentration range, but that range is too narrow to distinguish the
301 curve from its linear approximation at the observed scatter. This is a protocol
302 demonstration and limited distributional check, not evidence of an awakening
303 transition, a cortical field threshold, or a link between order and experience.
304 An informative test needs a broader concentration range and explicit treat-
305 ment of source mixing, finite samples and temporal pooling.

306 6 Discussion

307 The framework makes a conditional connection between thermodynamic in-
308 formation constraints, collective phase organization and compatible local rep-
309 resentations. Its formal contribution is to expose the assumptions needed to
310 connect those objects and to show that they can coexist in a common model.
311 Its biological contribution is a coupling-gated recovery hypothesis with de-
312 fined observables and failure conditions. The simulations are useful because
313 they restrict that hypothesis: finite ramps smear the stationary onset, an un-
314 pinned collective phase complicates independence diagnostics, and the tested
315 plasticity fails to recover environmental structure.

316 The proposal differs in emphasis from accounts based on integrated in-
317 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
318 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
319 candidate interaction medium and a mathematical compatibility condition.
320 It does not derive the exclusion principle of integrated information, identify
321 every broadcast with a field transition, or derive a generative model from a
322 dissipation bound. Continuous fields can carry incompatible local descrip-
323 tions, and discrete models can admit sheaf constructions. These facts prevent
324 continuity alone from settling the combination problem.

325 The formal hardware results are correspondingly limited. At matched
326 coupling resource, a fixed wiring support that excludes a better-correlated
327 pair can be suboptimal. On an atomless substrate, a kernel supported on
328 finitely many points contributes zero to the chosen continuum integral. The
329 latter is a property of that measure and observable; it does not imply that
330 digital hardware cannot approximate a field or support consciousness. No
331 biological-versus-artificial consciousness conclusion follows from these results.

332 The main empirical priorities are to estimate effective coupling and diffu-
333 sion in consistent units, test their proposed relation to extracellular geometry,
334 and measure recovery with an independently defined behavioral endpoint. A
335 favorable concentration-coherence relation would support a statistical de-
336 scription, not identify its physical cause. An unfavorable result in a regime
337 satisfying the stated assumptions would constrain that description. Likewise,
338 successful synchronization simulations cannot substitute for measurements

339 that distinguish ephaptic interactions from other coupling mechanisms.

340 The model does not establish that a globally consistent reflexive state
341 is conscious. That identification remains outside the conditional theorem.
342 Keeping it separate from the physical tests makes the proposal assessable:
343 the cortical coupling hypothesis and the recovery prediction can fail even
344 while the mathematical composition remains valid.

345 **Data and code availability**

346 The Lean development, simulation scripts, tests and saved numerical sum-
347 maries are maintained in the manuscript source repository. Publication sim-
348 ulation values are regenerated from saved summaries without rerunning pro-
349 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
350 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
351 in Supplemental Material. The source repository contains the reproducible
352 analysis rather than redistributing the raw recordings.

353 **Use of AI tools**

354 AI tools assisted simulation-code development and manuscript revision. Ma-
355 chine checking establishes the stated Lean results under their hypotheses;
356 it does not validate biological interpretations or replace scientific review of
357 generated code and prose.

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